

Market Segmentation on EV Market



1.Explain how and which ML model (algorithm) helped you in 2nd Project?

In the Project-2, two machine learning models are used to help with clustering and dimensionality reduction:

1. K-Means Clustering:

- What is K-Means?

K-Means is an unsupervised machine learning algorithm used for clustering data into distinct groups based on feature similarity. It works by partitioning the data into a pre-defined number of clusters, where each point belongs to the cluster with the nearest mean (centroid).

- How is K-Means used in the code?

The code first scales the extracted features like 'Two Wheeler', 'Three Wheeler', 'Four Wheeler', etc using StandardScaler to standardize the data. Then, it uses the Elbow Method by plotting the inertia, or within-cluster sum of squares, for different values of k to help determine the optimal number of clusters.

After determining the optimal number of clusters which is k=5 in this project, the code applies the K-Means algorithm to assign each state's sales data to one of the 5 clusters. This allows the states to be segmented into groups based on the similarity of their electric vehicle sales.

- Why K-Means?

K-Means is a popular algorithm for clustering due to its simplicity and efficiency. It works well when the goal is to group data points clusters based on similar characteristics. Since the goal here is to perform market segmentation by grouping states based on their electric vehicle sales, K-Means is a fitting choice.

2. PCA (Principal Component Analysis):

- What is PCA?

PCA is a dimensionality reduction technique that transforms the original dataset with many features into a smaller set of components, while retaining most important information. It helps visualize high-dimensional data by projecting it into a lower-dimensional space.

- How is PCA used in the code?

The code applies PCA after clustering with K-Means. It reduces the data to 2 principal components and then visualizes the clusters in a 2D scatter plot. Each point represents a state's electric vehicle sales, and the color represents the cluster the state belongs to.

- Why PCA?

PCA is used here mainly for visualization purposes. After performing clustering, the high-dimensional feature space (with several sales categories) is reduced to two dimensions, making it easier to see how the states are grouped into different clusters.

Some Libraries used in this project

pandas:

- Used for data manipulation and analysis.
- Helps in reading the CSV file, handling missing data (using fillna()), and manipulating data frames.

numpy:

- Provides support for numerical operations and working with arrays.
- Used to handle numerical calculations and create ranges of values.

matplotlib:

- Used for data visualization.
- Helps in creating bar plots and scatter plots to visualize sales by states, clusters, and vehicle types.

2. Elaborate on the final conclusion & insights gained from the research/analysis work.

The analysis performed in the code provides several important insights into the electric vehicle market across different states, allowing for effective market segmentation and identifying key trends.

The analysis provided valuable market segmentation by clustering states based on their electric vehicle sales data across categories like two-wheelers, four-wheelers, goods vehicles, and more. The K-Means clustering algorithm grouped states into five clusters, each with distinct sales patterns. This segmentation highlights states with similar market characteristics, enabling targeted strategies for marketing and infrastructure development. Clusters with high sales, such as cluster 1, represent key regions for further investment, while clusters with lower sales, like cluster 0, indicate potential growth areas with untapped markets.

Visualization through PCA made it easier to understand how states relate to each other within clusters. Furthermore, the analysis identified specific states excelling in various EV categories, offering insight into where marketing and sales efforts should be focused. Overall, the research provides actionable insights for expanding EV markets, targeting specific regions with tailored policies, and developing infrastructure to support future EV growth, ensuring a more effective promotion of electric vehicles across different states.

3. How will you improve upon the Market Segmentation Project given additional time & some budget to purchase data? (in terms of Datasets collection - name what columns points you will search for & what additional ML models you would like to try)

To refine the segmentation and gain deeper insights into the EV market, I would search for datasets that provide more granular information, including:

1. Demographic Data:

- Income Levels: EV adoption tends to correlate with income, so state or district-wise average income would help segment markets based on affordability.
- Population Density: Densely populated areas may require more public EV infrastructure, while rural areas might focus on individual ownership.
- Education Levels: Awareness of environmental impact or new technologies often correlates with education, influencing EV adoption rates.

- Urban vs Rural Population: Understanding the proportion of urban vs rural populations can inform infrastructure needs and demand for different EV types.
2. Economic and Infrastructure Data:
- EV Charging Stations per State: The number and location of charging stations can heavily influence EV adoption and would help in identifying infrastructure needs.
 - Electricity Prices: Lower electricity prices can be an incentive for EV adoption, so tracking average prices per state could provide useful insights.
 - Government Incentives: Availability and scope of state or district-level EV incentives, subsidies, or tax breaks, as this would greatly impact sales.
3. Vehicle-Specific Data:
- Fuel Prices: Comparing traditional fuel prices to electricity prices could explain why certain states might prefer EVs over conventional vehicles.
 - EV Ownership Costs: Including data on EV purchase costs and maintenance expenses could help in assessing regions where people are more likely to adopt EVs.
 - EV Sales Over Time: Time-series data would provide insights into the growth of EV adoption, revealing trends and changes in market dynamics.

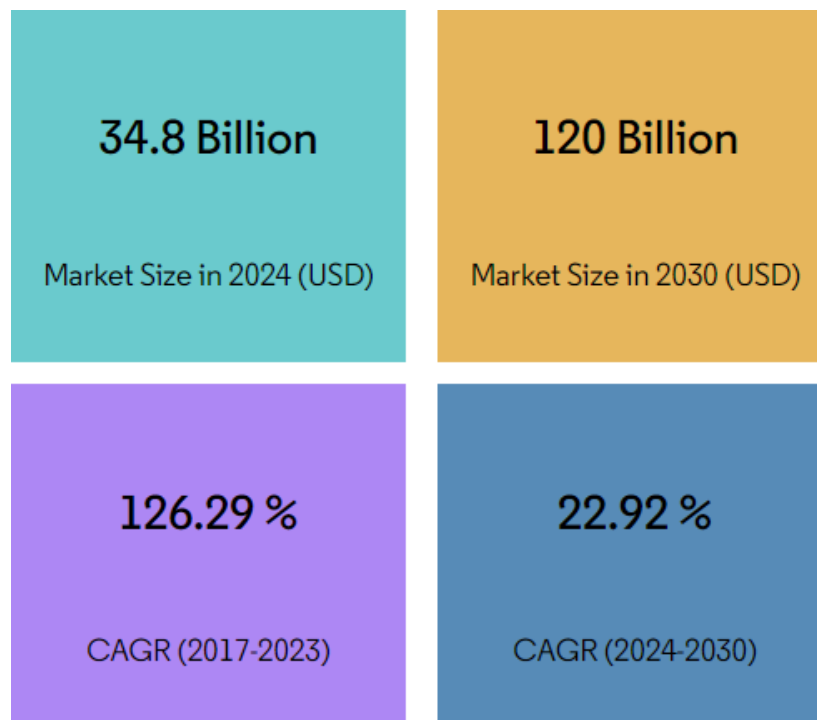
Additional Machine Learning Models:

1. **Gaussian Mixture Models (GMM):**
- Unlike K-Means, which assigns hard clusters, GMM can assign probabilities to each data point belonging to a cluster. This would allow for softer segmentation, where states that are borderline between two clusters could be identified as potential transition regions.
2. **Random Forest Classifier for Predictive Analysis:**
- A predictive model could be trained using Random Forest to predict the likelihood of EV adoption in different states based on factors like income, infrastructure, and government incentives. This would provide actionable forecasts for potential market expansion.

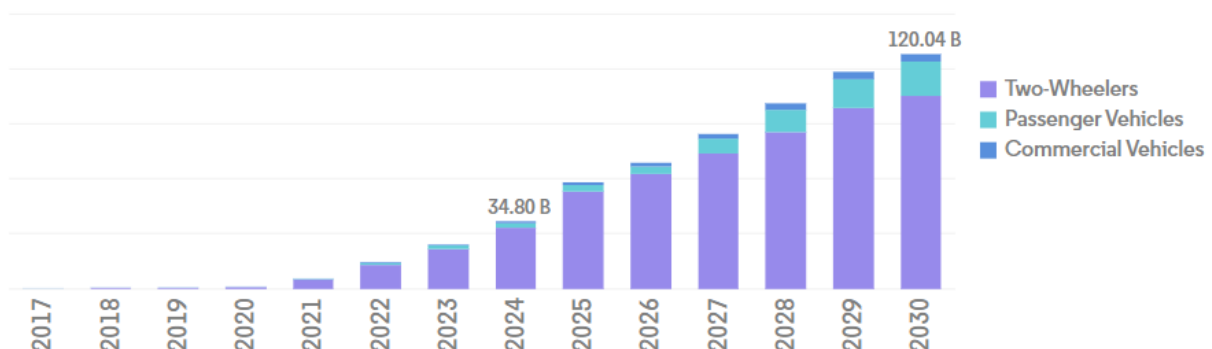
4.What is the estimated Market Size for your Market Domain (non-segmented) in Numbers?

India Electric Vehicle Market Analysis

The India Electric Vehicle Market size is estimated at 34.8 billion USD in 2024, and is expected to reach 120 billion USD by 2030, growing at a CAGR of 22.92% during the forecast period (2024-2030).



India Electric Vehicle Market, BY vehicle type, By Value In USD, 2017 - 2030



5.Name top 4 Variables/features which can be used to create most optimal Market Segments for your Market Domain

Income Levels:

- Higher income levels generally correlate with a greater ability to afford electric vehicles, which tend to have higher upfront costs compared to traditional vehicles. Understanding the income distribution in different states can help identify target segments that are more likely to adopt EVs.

Number of EV Charging Stations in all states:

- The availability and accessibility of charging infrastructure directly influence EV adoption. States with a higher density of charging stations are more likely to see increased EV sales, making this a key factor in market segmentation.

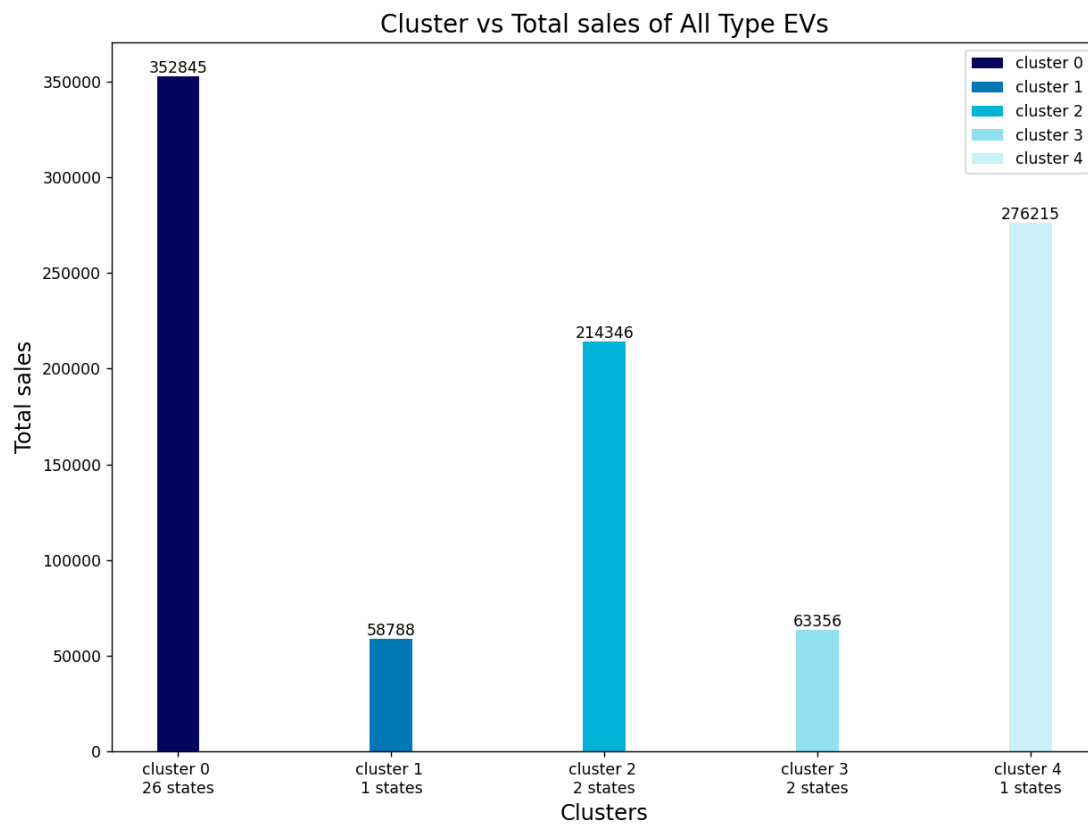
Fuel Prices (Petrol/Diesel/CNG):

- The fuel prices are increasing day-by-day. Which is beneficial for electric vehicles market as higher prices for traditional fuels can drive consumers towards electric vehicles, making this a crucial variable in market segmentation.

Environmental Awareness and Attitudes:

- Environmental awareness and attitudes refer to the general public's understanding of and concern for environmental issues, particularly those related to climate change, pollution, and sustainability. This variable plays a significant role in influencing consumer behaviour and decision-making regarding electric vehicles.

Segmentation Results:



	Two Wheeler	Three Wheeler	Four Wheeler	Goods Vehicles	Public Service Vehicle	Other Vehicles
cluster						
1	51149	6155	2	30	851	601
2	71467	129309	10263	202	83	3022
3	50726	9811	130	2358	38	293
4	18295	257159	368	53	327	13

Cluster 2 has Maximum sales of Two Wheeler which is 71467.
It has states: ['Delhi' 'Karnataka'] which are best to sell Two Wheeler.

Cluster 4 has Maximum sales of Three Wheeler which is 257159.
It has state: ['Uttar Pradesh'] which is best to sell Three Wheeler.

Cluster 2 has Maximum sales of Four Wheeler which is 10263.
It has states: ['Delhi' 'Karnataka'] which are best to sell Four Wheeler.

Cluster 3 has Maximum sales of Goods Vehicles which is 2358.
It has states: ['Chhattisgarh' 'Tamil Nadu'] which are best to sell Goods Vehicles.

Cluster 1 has Maximum sales of Public Service Vehicle which is 851.
It has state: ['Maharashtra'] which is best to sell Public Service Vehicle.

Cluster 2 has Maximum sales of Other Vehicles which is 3022.
It has states: ['Delhi' 'Karnataka'] which are best to sell Other Vehicles.

GitHub Link: [Project-2 Link](#)